23 April 2026 10:38

API

```
# post request:-
# import requests
# url='https://jsonplaceholder.typicode.com/posts'
# data={
# 'title':'test',
# 'body':'adding for checklist',
# 'userId':101
# }
# response=requests.post(url,data)
# #check if the request was successful(HTTP satatus code 201 fro created )
# if response.status_code==201:
#     print(f'success , new post created with {response.json()["id"]}')
# else:
#     print(f'ERROR {response.status_code}')

#put and delete
# import requests
# #define the URL and Updated Data
# url='https://jsonplaceholder.typicode.com/posts'
# data={
# 'id':92,
# 'title':'id minus libero illum nam ad officiis',
# 'body':'earum voluptatem facere provident blanditiis velit laboriosam\npariatur accusamus odio saepe\ncumque dolor
qui a dicta ab doloribus consequatur omnis\ncorporis cupiditate eaque assumenda ad nesciunt',
# }
# #make  a PUT request to update the resource
# response=requests.put(url,data)
# #check if the request was successful(HTTP satatus code 200 fro created )
# if response.status_code==200:
#     print(f'success ,  post updated {response.json()}')
# else:
#     print(f'ERROR {response.status_code}')
# #make a DELETE request to delete the resource
# response=requests.delete(url)
# #check if the DELETE request was succesfull (200 for ok)
# if response.status_code==200:
#     print("success post deleted ")
# else:
#     print(f'ERROR {response.status_code}')
#weather report generation
# import requests
# api_key="d7f0a9c19c0a348daed92969e0565afe"
# city="Bangalore"
# # url=f'https://api.openweathermap.org/data/2.5/weather?q={city}&appid={api_key}&units=metric'
# url=f'https://wttr.in/{city}?format=j1'
# response=requests.get(url)
# # print(response.status_code)
# print(response.json()["current_condition"][0]["temp_C"])
# import requests
# api_key="d7f0a9c19c0a348daed92969e0565afe"
# city="Chennai"
# url=f'https://api.openweathermap.org/data/2.5/weather?q={city}&appid={api_key}&units=metric'
# response=requests.get(url)
# # print(response.status_code)
# data=response.json()
# print('City :',data['name'])
# print('Temparature :' ,data['main']['temp'],'C')
# print('weather :',data['weather'][0]['description'])
# print('Humidity',data["main"]["humidity"],"%")
# print("wind Speed",data['wind']['speed'],'m/s')
# print("timezone",data['timezone'])
```

Async Processing:-

Asynchronous programing lets one program handle multiple waiting tasks without blocking .

Instead of :-
Connect to server A --->wait
Then Server B -->waiting
Then Server C --> waiting
Anys does:
Start A
Start B
Start C
Continue when each fishes :

Sequential:- 100 servers.. * 2sec = 200 sec
Asyn concurrent :- 2-10seconds depending on limits
Concurrency operation:-

Method	Best for	Devops Use
Asyncio	Network/I/o	Best overall
Threading	Legacy lib	Parallel request
Multiprocessing	Large system	Scheduling jobs

Define Async function

```
Async def deploy():
        Pass
```

Await --> pause current task until result is ready
Await api_call()

Loops the task concurrently
Engine that run task
Asyncio.run(main())
Gather:-
Run multiple task
Await asyncio.gather(taask1(),task2()....)

AI Modules:--

AI artificial intelligence , machine performing tasks requiring human intelligence
Examples:-
Chat boards
Fraud detection
Face recognition

ML(machine learning)-;
Machine learns patterns from data..
Input:- CPU USAGE , MEMORY USAGE--> PREICT THE SERVERR FAILURES
USED IN DEVOPS;-
PREDICT OUTAGES
DETECT ANOMALIES
CAPACITY PLANNIG
DeepLearning :-use neural networks with many layers;-
Example-
Speech recognition
Vision systems
Large Language Models
What is NLP (natural Language processing =computers understanding text)
Example:-sentiment analysis , translation , chatbots
LLM(large language Model) trained on huge text data
Exe-chatgpt
Claude

Gmini
Mistral
Llms generate next probable tokes
User-how to restart nginx server?
LLm predicts likely a next word :
Use systemctl restart nginx
It doesn't know like human -- it predicts based on the training

Token:- words are split into chunks
Example;-
Deploy kubernates cluster
May be become 4-6 tokens
Embedding:- text converted in to vectors for search
RAG used in :-

Devops use case of LLMs:-

- **Auto generate terraform**
- **Kubernates troubleshooting**
- **Incident RCA (root cause analysis)**
- **Log summarization**
- **Ci/cd pipelines**
- **Security checks**